DEEP LEARNING

Convolutional Neural Networks Visual Guide

How CNNs see:
convolutions, pooling, receptive fields, skip connections.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Convolution Operation

A small learned kernel slides across the image.

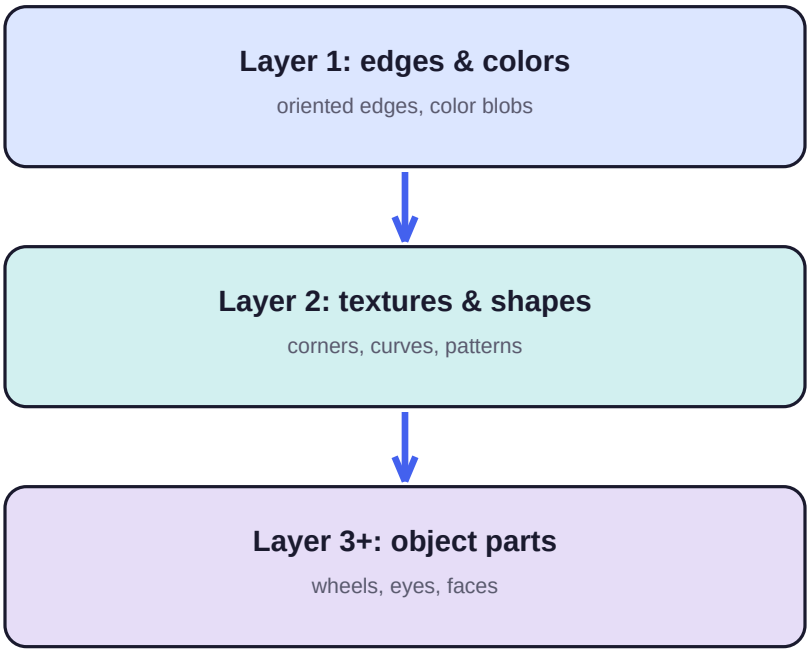
At each position the kernel computes a weighted sum. One kernel = one feature detector.

Weight sharing: the same kernel is reused everywhere — far fewer parameters than a dense layer.

STEP 2/6

Filters Learn a Hierarchy

Simple features compose into complex ones.



Visualize the filters to audit what the network 'sees'. Early layers are generic; late layers are task-specific.

STEP 3/6

Pooling: Shrink and Summarize

Keep the strongest signal, drop the resolution.

Pooling buys translation invariance and cuts compute. Modern nets often use strided convolutions instead of pooling.

STEP 4/6

Receptive Fields Grow

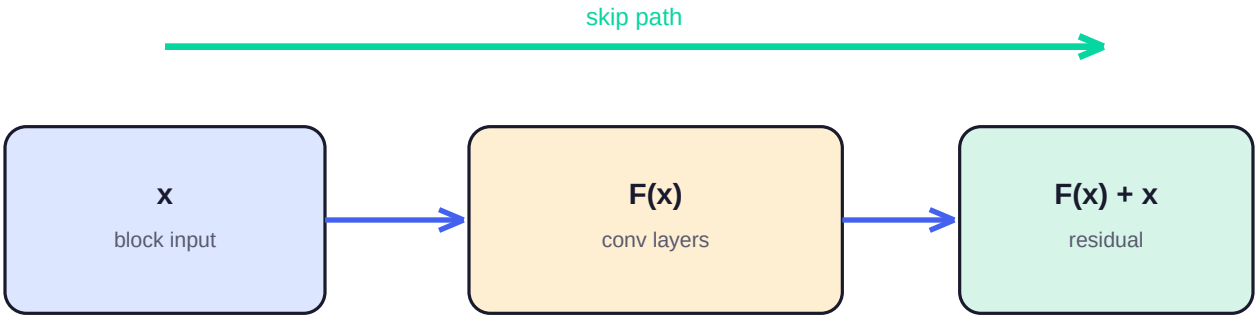
Deeper neurons see more of the image.

The receptive field is why deep CNNs understand context: late neurons integrate the whole scene.

STEP 5/6

Skip Connections (ResNet)

A shortcut lets gradients flow through 100+ layers.



The network learns the residual F(x) instead of the full mapping — much easier to optimize.

Residual nets made very deep CNNs trainable and are the backbone of most vision systems.

STEP 6/6

CNN Design Rules

Defaults that work.

- 3x3 kernels, stacked deep, beat one big kernel: same receptive field, fewer params, more nonlinearity.
- Stride 2 (or pooling) halves resolution each stage while doubling channels: 64 -> 128 -> 256 -> 512.
- Batch normalization after convolutions stabilizes training and allows higher learning rates.
- Data augmentation (crops, flips, color jitter) is free training data — almost always worth it.
- For a new vision task, fine-tune a pretrained ResNet/EfficientNet before training from scratch.

STEP ✓/✓

Key Takeaways

CNNs — Remember

- Convolution = learned kernel sliding over the image; weight sharing keeps params low.
- Filters form a hierarchy: edges -> textures -> object parts.
- Pooling/striding downsamples for invariance and efficiency; receptive fields grow with depth.
- Skip connections (ResNet) let gradients flow through very deep networks.
- Start from a pretrained backbone and fine-tune; augment aggressively.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev